137.5 MHz High-Dynamic-Range RF Front-End

Design & Implementation of a VHF Satellite Ground Station LNA Stage

Don't worry for the 16 pages, if you just want to see the summary go *here*

Project Overview

- ***Target:*** *137.5 MHz VHF Downlink*
- ***Active Device:*** *PGA-103+ E-PHEMT MMIC*
- ***Out-of-band rejection (88-108 MHz):*** *>12.5 dB*
- ***Insertion Loss (137 MHz):*** *1.09 dB*
- ***Stability:*** *Compensated unconditionally stable at low freq (R+L shunt)*
- ***PCB Layout:*** *CPGW (50Ω), via-stitching*

This project covers the design, simulation, and physical implementation of an active RF front-end optimized for receiving meteorological satellite downlinks in the **137 MHz VHF band** (specifically NOAA-15/18/19 and Meteor-M series). The primary goal is to serve as the low-noise receiver core for a homebrew, satellite ground station.

While receiving 137 MHz signals on paper seems straightforward, operating in a dense urban environment like Barcelona presents a severe RF challenge. The local spectrum is heavily saturated by high-power commercial FM broadcast transmitters (88–108 MHz) operating just a few megahertz below our target frequency band. Standard low-cost LNAs or wideband receivers easily get driven into 1-dB compression or suffer from intermodulation distortion when exposed to these strong out-of-band signals.

A quick note before getting into the schematics:
I built this project to get my hands dirty with real hardware. I had not yet taken a formal RF course when I started it, so this project was also an opportunity to learn RF engineering through a practical hands-on challenge. This seemed like a good way to get my first real experience in the field.

Objectives and component selection:

The primary objective of this stage is to amplify and filter weak satellite signals received from the antenna while introducing the **minimum thermal noise** to the receiving chain, as system sensitivity is highly dependent on the first active stage (according to Friis' formula).

To guide my design process, I established the following technical objectives and target specifications:

- **Operating Frequency:** Centered at 137.5 MHz, specifically targeting the VHF downlink band.
- **Noise Figure (NF):** Optimized to preserve the system's SNR, targeting an overall cascaded NF below 1.5-2dB (based on my initial Friis calculations).
- **Linearity and Dynamic Range:** A high OIP3 to prevent the active stage from being driven into 1-dB compression by strong adjacent-band signals. (88-108 MHz FM broadcasts).

For the active device, I ultimately decided to use the **PGA-103+ E-PHEMT MMIC** from Mini-Circuits. While I initially considered other LNAs, I realized I needed a device with an exceptionally high dynamic range (OIP3 of roughly +38 dBm). This robust linearity is a critical requirement for my front-end architecture, as it allows the LNA to withstand high-power out-of-band interference from city FM transmitters without saturating.

1. RF Environment & Link Budget Analysis

Before designing any stage of the RF front-end, I first wanted to estimate the signal levels that my receiver could realistically encounter. Rather than choosing the filter attenuation or LNA characteristics arbitrarily, I used a simple link budget analysis to estimate the maximum FM power that could reach the antenna under worst-case conditions. From these results, I derived the minimum rejection required from the first stage to keep the LNA operating within its linear region avoiding saturation or intermodulation products.

1.1 Assumptions

As I said earlier, Barcelona is heavily saturated by high-power FM transmitters, the closest and strongest one (Torre de Collserola) is around 4.3km away from my antenna. This telecommunication tower emits roughly at 240kW in the FM band.

1.2 Maximum Received Power (Worst-Case Scenario)

To mathematically justify the filter requirements, I evaluated the worst-case scenario: assuming clear line-of-sight to the Collserola tower with zero structural attenuation from nearby buildings. With the tower transmitting at 240 kW, the Equivalent Isotropically Radiated Power (EIRP) is approximately $+\mathbf{83.8}$ **dBm.**

Using the Free Space Path Loss (FSPL) equation for a distance of $d = 4.3$ km and a frequency of $f = 100$ MHz:

$$FSPL = 20\log_{10}(d) + 20\log_{10}(f) + 32.44$$

$$FSPL = 20\log_{10}(4.3) + 20\log_{10}(100) + 32.44 = \mathbf{85.11\ dB}$$

Assuming a worst-case antenna gain of 0 dBi in the FM band, the total received power at the antenna terminals is:

$$P_{rx} = EIRP - FSPL + G_{rx}$$

$$P_{rx} = 83.8\text{ dBm} - 85.11\text{ dB} + 0\text{ dBi} = \mathbf{-1.31\ dBm}$$

1.3 LNA Saturation & Minimum Filter Attenuation

To find the minimum attenuation required, I analyzed the PGA-103+ operating limits. According to the datasheet, at 0.05 GHz and a $+5$ V supply, the amplifier provides a typical gain of 26.5 dB and an output 1 dB compression point (OP_{1dB}) of $+20.0$ dBm. This puts the input compression point (IP_{1dB}) at:

$$IP_{1dB} = OP_{1dB} - G = 20.0\text{ dBm} - 26.5\text{ dB} = -6.5\text{ dBm}$$

(this equation uses “G-1” instead of only G due to the nature of the compression itself, I will be using the G alone to be conservative)

Therefore, to purely prevent LNA saturation, the filter must provide a minimum attenuation (A_{min}) of:

$$A_{min} = P_{rx} - IP_{1dB} = -1.31\text{ dBm} - (-6.5\text{ dBm}) = \mathbf{5.19\text{ dB}}$$

While 5.19 dB is the strict minimum to avoid the 1 dB compression point, operating an LNA on the edge of its non-linear region severely degrades the noise figure and creates a vulnerable system. To maintain a robust safety margin, I estimated a minimum target attenuation of **10 dB** and an optimal target of 15 dB.

1.4 Intermodulation Distortion (IMD) Verification

Even with the LNA safely in its linear region, I needed to verify if intermodulation products, specifically the 5th order product (IMD5), would bleed into my 137 MHz band and degrade the SNR.

Note: The maximum IMD3 product sits at (2·108MHz-88MHz = 128MHz). IMD3 won't be a problem.

Based on the PGA-103+ datasheet, the output IP3 (OIP_3) at 0.05 GHz is $+36.7$ dBm.

Subtracting the 26.5 dB gain, the input IP3 (IIP_3) is $+10.2$ dBm. For a conservative first-order estimate, I approximated IIP5 as IIP3 + 10 dB.

Evaluating an intermediate nominal attenuation of 12.5 dB from my Notch filter, the FM signal reaching the LNA (P_{in_LNA}) drops to -13.81 dBm (-1.31 dBm $-$ 12.5 dB). The IMD5 power generated at the input is calculated as:

$$P_{IMD5} = 5 \cdot P_{in\_LNA} - 4 \cdot IIP_5$$

$$P_{IMD5} = 5(-13.81) - 4(20.2) = -69.05 - 80.8 = \mathbf{-149.85\text{ dBm}}$$

The thermal noise floor for the ~ 150 kHz bandwidth of the Meteor-M LRPT signal is around -122 dBm. With the IMD5 product sitting at -149.85 dBm, it remains nearly 28 dB below the noise floor. This confirmed that aiming for a $10 - 15$ dB rejection range is perfectly sufficient.

2. System Architecture & Cascaded Noise Analysis for the 1st stage

Having defined the environmental constraints and the required filter rejection, one of the main problems I encountered while designing this first stage was how to maintain a minimum NF while at the same time having to block the FM transmissions.

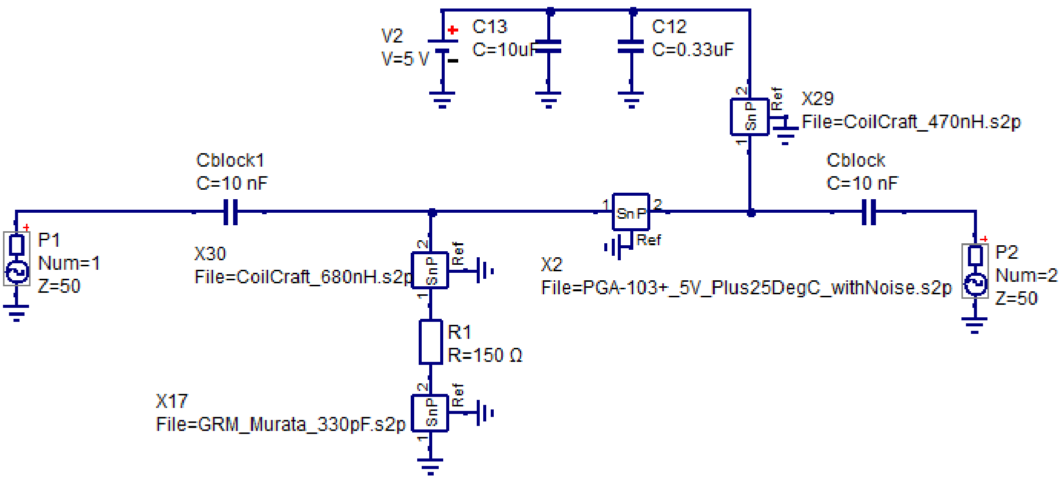
According to Friis's formula, any passive loss introduced prior to the first amplification stage translates to a direct, **1-to-1** degradation of the total system Noise Figure. Placing a passive bandpass or notch filter directly at the antenna input to block commercial FM transmissions introduces a significant value of insertion loss.

At first, the MMIC I used allowed me to consider not implementing any filter at the input thanks to its high dynamic range. However, as demonstrated in the previous link budget, the sheer power of the nearby FM transmitters made a Notch filter strictly necessary. The FM band (88-108MHz) is critically close to my operating frequency, meaning I needed a filter to bring around 10-15dB attenuation at 88-108MHz and have almost no effect at 137MHz.

To mitigate its effect and NF degradation, I used discrete components with high Q and high SRF; specifically, I used the Murata GRM series for the capacitors and the CoilCraft 0603CS/0805LS series for the inductors (LS because of the low DCR).

According to the datasheet, the PGA-103+ has a NF of 0.6dB at 1GHz (which interpolates to around 0.45-0.5dB at 137MHz) and is optimized to bring good output and return loss for a range of frequencies without the need to add external matching components. This means you don't need to adapt the LNA to have a good NF and optimal S values. This is true when you're only measuring the LNA in an optimal environment, without the low frequency stabilization shunt and the whole *bias tee* circuit.

The following circuit shows how I simulated the real parameters of the LNA between two 50 Ω ports:



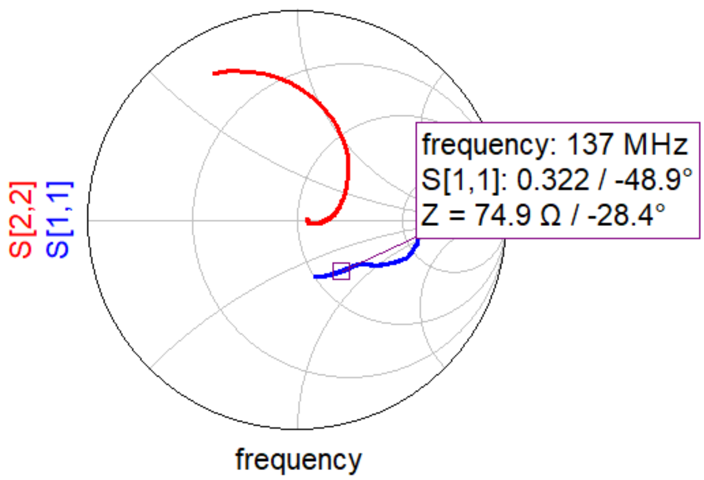
Circuit taken directly from the datasheet manufacturer, with a low frequency stabilization shunt, both DC block capacitors, L_{choke} *to isolate the RF from the power supply and the filtering capacitors next to the voltage source, using .s2p files for the critical components.*

f	dB(S[1,1])	dB(S[2,1])	dB(S[2,2])	NF
134	-9.82	25.5	-25.1	0.597
135	-9.83	25.4	-25.2	0.595
136	-9.83	25.4	-25.2	0.594
137	-9.84	25.4	-25.2	0.593

Although the values are not what the datasheet provided, we still have an excellent **0.593dB** for the NF, -**25.2dB** for the S22 and **25.4dB** for the S21 measures despite having a relatively low value of **-9.84dB** for S11, even though, according to the manufacturer in the *datasheet* it says that this value is optimal for this LNA.

[***important:*** this NF is not true; in the end it will be slightly higher (0.732) I will add a shunt-to-ground R+L and a pi-resistors-network for stability as you will see shortly.]

The -9.84dB value for S11 represents the input return loss, meaning that the LNA does not have an impedance of 50Ω in its input pin at our operating frequency (137MHz), and a considerable amount of energy is reflected by its entrance due to this mismatch.



You can see it in the figure; It has an impedance of $\boldsymbol{Z = 65.88 - j35.62\ \Omega}$ (moderately capacitive impedance).

This actually makes all the sense; The minimum noise point (Γ_{opt}) almost never coincides with the maximum power transfer point (the conjugate of Zin). Matching to a pure 50 Ω maximizes gain and S11, but it moves you away from Γ_{opt} which is why the NF skyrockets when you try to match it perfectly. What we want is a "**noise-matched compromise**". As I said before, in the datasheet there's a comment saying that the value of -8dB/-11dB is considered optimal for this LNA, and this is why it does not need any adapting networks.

If we try to completely or partially adapt this stage, we will need to add passive components before the LNA, and because of Friis's (1-to-1 degradation) we will introduce an unacceptable value of NF without even implementing the filter yet, so I decided to not over-engineer this previous stage.

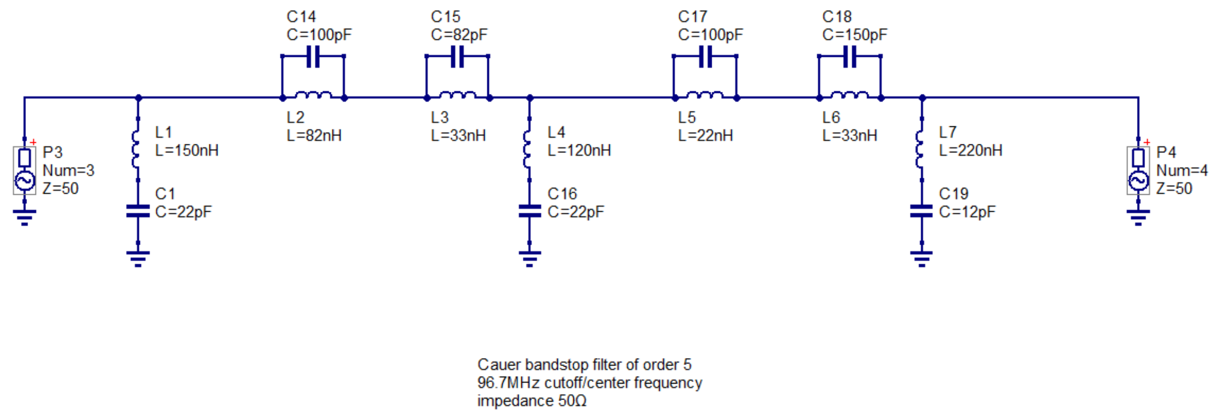
Moving on with the design and mentioning Friis's again, the first stage is critical in RF Front-Ends, it practically determines the **overall NF** you will have in the total system. Since the LNA provides an exceptional gain, it mitigates the NF contributions of subsequent stages.

As I mentioned before, we need a filter to bring around 10-15dB attenuation at 88-108MHz and have almost no effect at 137MHz. These are specifications a Cauer can perfectly cover.

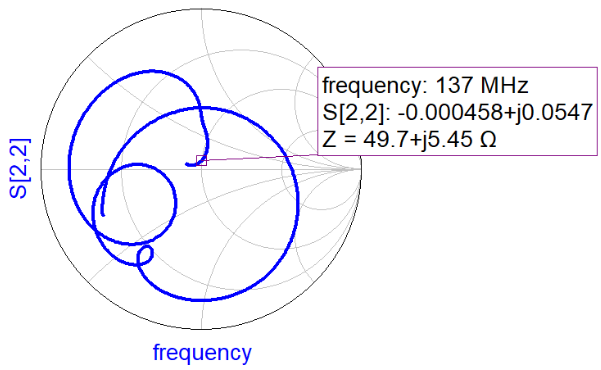
While I considered different topologies and orders for the filter, I ultimately decided to implement a 5th order Cauer-Notch filter and achieved an IL, and therefore a NF, of about **1.09dB** at 137MHz.

For this simulation I will be using the s2p files from the manufacturers, the poles and zeros as well as the output impedance will be slightly displaced from the theoretical 50 Ω.

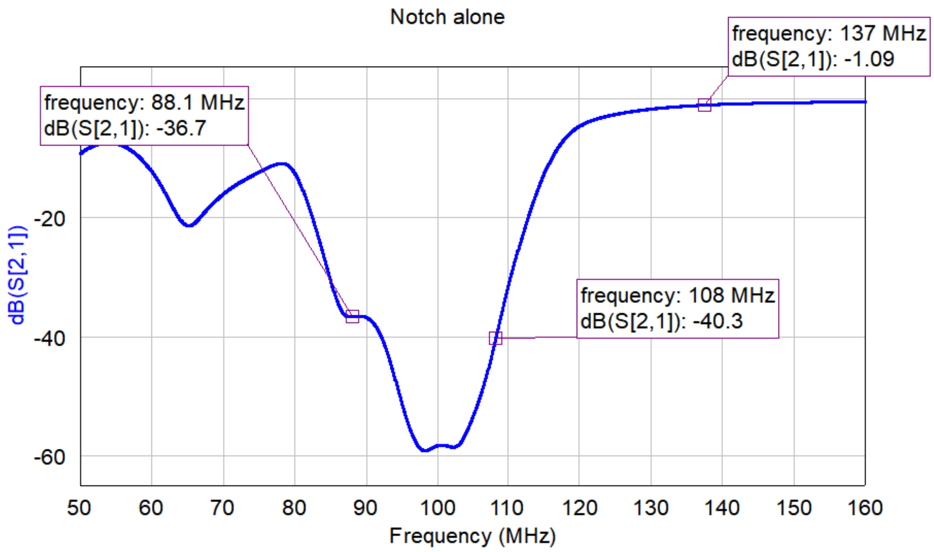
The following figures show the S-parameters and the rejection in the FM band:



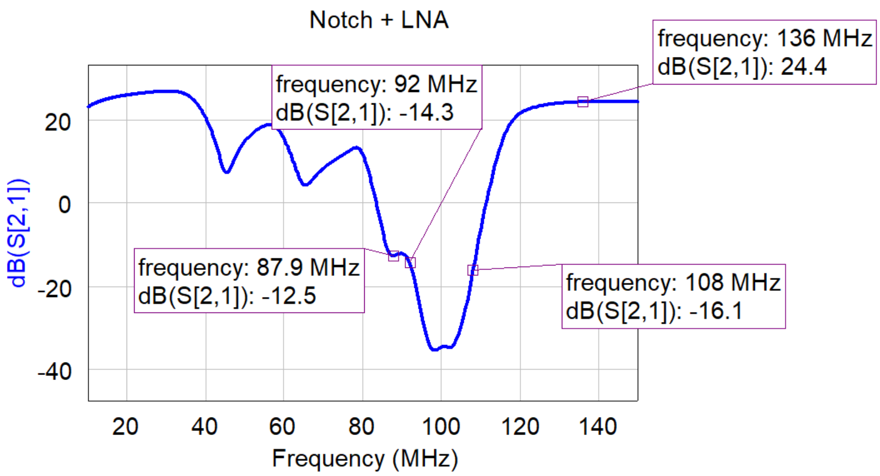
Circuit design of the filter using the tool: "filter synthesis" from QUCS-S, the components have been changed for the .s2p files when simulating.



Output impedance of the filter, we can see that it is slightly inductive.



Graphic with the values of S21 at different frequencies covering all the FM band showing an exceptional attenuation and the IL at 137MHz being 1.09dB.



Graphic showing S21 values implementing the filter in cascade with the LNA

f	dB(S[1,1])	dB(S[2,2])	dB(S[2,1])
135	-13.6	-22.3	24.4
136	-13.2	-22.5	24.4
137	-12.8	-22.6	24.4
138	-12.5	-22.6	24.5

As you can see, the attenuation across the entire FM band is strictly under -10 dB, meeting the specifications while introducing only -1.09 dB of insertion loss (IL), a perfectly justified number given the specifications of a 5th order notch filter.

Regarding our operating frequency, we can notice the S11 value has improved with respect to the LNA alone. Figure 3 shows the output impedance of the filter is inductive, meaning that without really looking for it, acts as a partial matching-network between the filter itself and the LNA providing a better S11 value.

Until now I focused only on the results without taking into account critical aspects in any RF design: its stability across all the operating frequencies the LNA is functional (in our case 50MHz to 4GHz) and the correct operation of the power supply.

If the voltage that arrives at the LNA drops below 4.74V it can significantly degrade its gain and NF so we must make sure to use optimal components with low DCR to provide a constant and sufficient voltage to the PGA-103+.

3. Stability & DC analysis

In any RF design, especially when using LNAs, it is essential to verify and guarantee circuit stability across all operating frequencies. Typically, the 1 MHz to 50 MHz band is the most problematic, depending on the device. At these frequencies, LNAs exhibit exceptional gain, meaning any input signal can cause the circuit to oscillate unconditionally.

Stability can be calculated using different expressions and factors. The two most common methods involve **Rollett's stability factor** (K) combined with the **delta determinant (Δ)** and the **μ-factor (geometric stability factor)**.

For a circuit to be unconditionally stable using the Rollett criterion, it must satisfy both **K > 1** and **Δ < 1** simultaneously.

Alternatively, the **μ-factor** simplifies this assessment by providing a single, sufficient metric where unconditional stability is guaranteed if **μ >1**. While the K-factor only indicates whether a device is stable, the **μ**-factor offers the advantage of quantifying the exact degree of geometric stability at the input or output.

We can quantify these factors using the following expressions:

Using the Rollet criterion:

$$K = \frac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|} \qquad \Delta = S_{11}S_{22} - S_{12}S_{21}$$

Condition for unconditional stability: K > 1 and Δ < 1.

Using the **μ**-factor:

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

Condition for unconditional stability: $\mu > 1$

From these formulas, we can easily identify how strongly stability depends on the system gain (S21). These expressions highlight a fundamental trade-off: as gain increases, the circuit inherently becomes more prone to instability.

These are the mathematical conditions required for a circuit to achieve ideal stability. However, in real-world designs, we must account for various non-idealities, such as temperature fluctuations, component tolerances, and PCB fabrication variances.

To guarantee that the circuit remains safely within the unconditionally stable zone under all operating conditions, I implemented a design margin. Instead of designing strictly for the boundary limits ($K = 1$), ($\mu =1$), I will be aiming for a K-factor between 1.2 and 1.5, and a μ-factor between 1.1 and 1.2.

Since I am using QUCS-S for the circuit simulation, there is no need to compute these expressions manually, as the software natively embeds tools to calculate and plot both factors, I will also plot the K and μ vs *f* to see at what frequencies the values drop below 1.

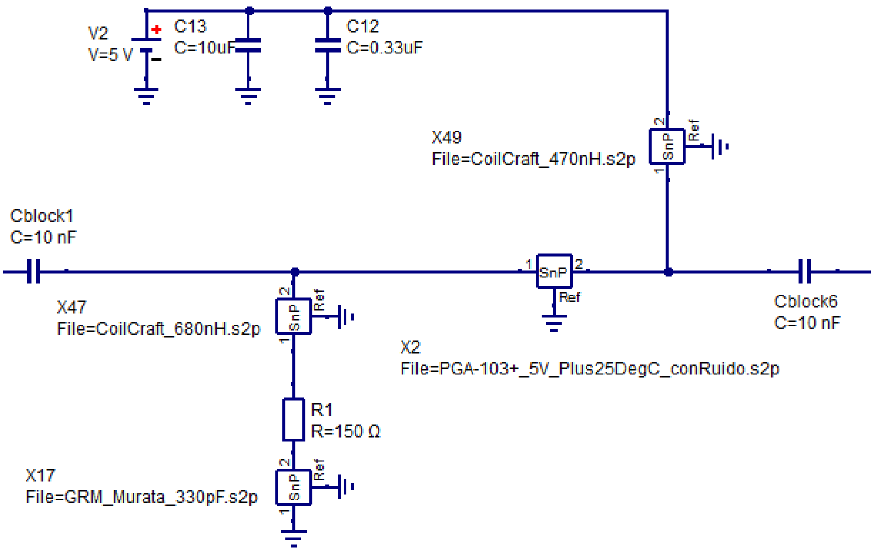
Important Notes:

For this simulation I'm using .s2p files and parasitic models taken directly from each manufacturer's datasheet, my frequency limit 500MHz using the first order parasitic model. **Beyond the stated limit, what you're seeing isn't a real measurement anymore**, it's the simulator extrapolating, I treat that as indicative, not reliable. For my 470nH and 680nH models I will be using the parasitic model with DCR+LC tank beyond 240MHz.

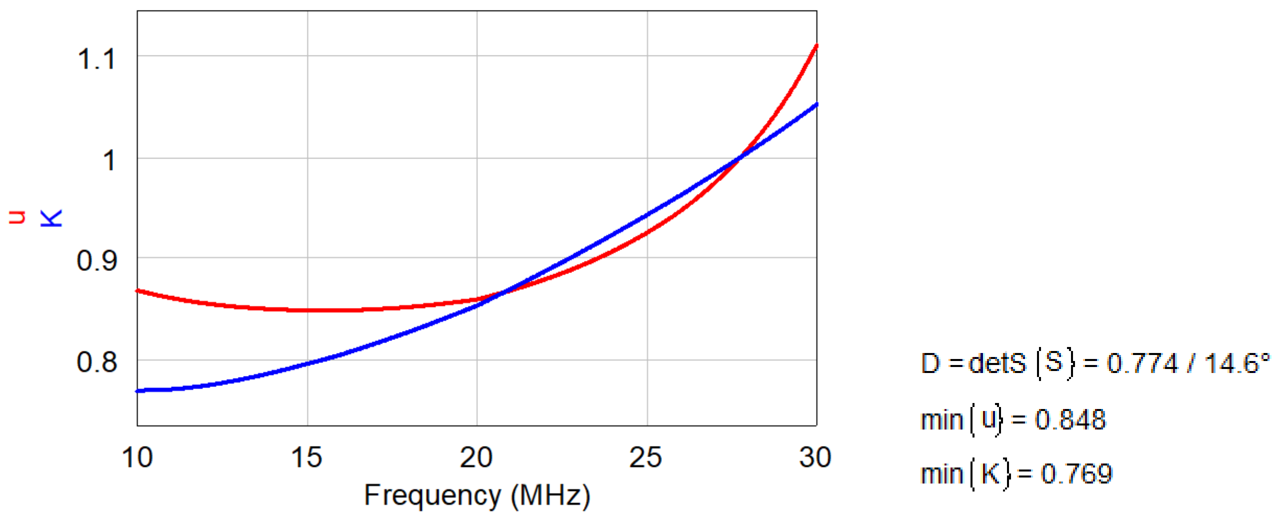
Beyond the first resonant mode (> 500 MHz), the parasitic model no longer captures the distributed effects of the packaging, making the stability curve at 4GHz only an ideal approximation. However, at these frequencies, system stability is guaranteed by two factors:

- According to the manufacturer, the internal architecture of the PGA-103+ E-PHEMT is already unconditionally stable above 200 MHz.
- The dominant reactance of the stabilization inductors acts as a high-impedance isolation, preventing the amplifier's input from being loaded and preserving its native stability.

The following figures show the 2 circuits I used to get the stability factors:



Circuit using s2p files and simulating from 10MHz to 240MHz.



We can see that both factors drop below 1 at low frequencies; this is expected, since low frequencies are exactly where LNAs tend to become potentially unstable. One of the most common techniques to guarantee stability at low frequencies is to use a R+L shunt-to-ground.

This branch acts as a dynamic impedance: at low frequencies the inductor behaves as a short circuit, "bleeding" the low frequencies to ground, while at high frequencies the combination of both presents a large enough impedance to behave as an open circuit.

I've sized this branch so that at 137 MHz the branch already presents a high impedance and doesn't load the LNA.

I used a value of 50 Ω for the R and 470nH for the inductor, if we calculate the impedance at 10MHz and 137MHz we have:

$$Z = R + j\omega L$$

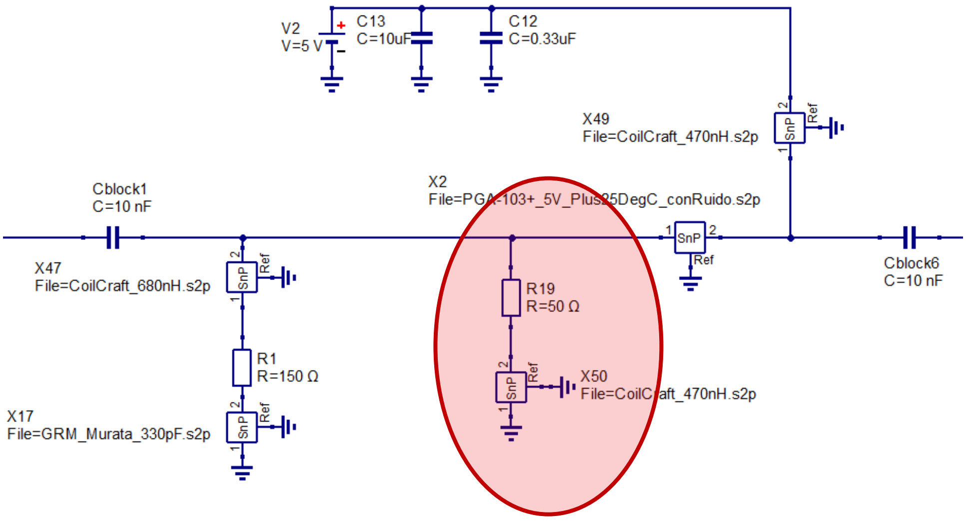
At 10 MHz: $\omega L = 2\pi \cdot 10 \times 10^{6} \cdot 470 \times 10^{-9} \approx 29.5\ \Omega$

$$Z_{10MHz} = 50 + j29.5\ \Omega \ \rightarrow\ |\boldsymbol{Z_{10MHz}}| \approx \mathbf{58\ \Omega}$$

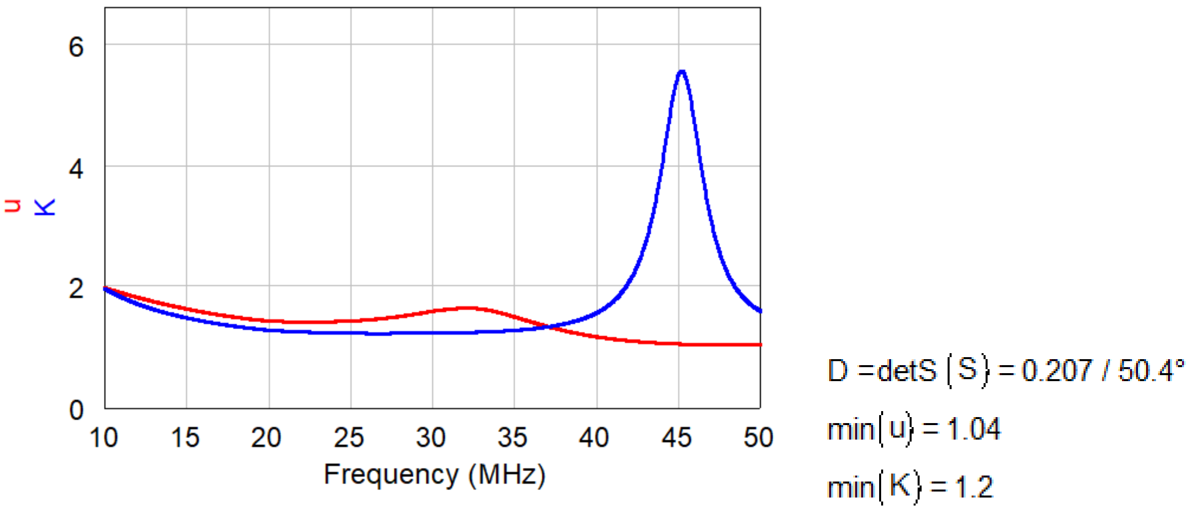
At 137 MHz: $\omega L = 2\pi \cdot 137 \times 10^{6} \cdot 470 \times 10^{-9} \approx 405\ \Omega$

$$Z_{137MHz} = 50 + j405\ \Omega \ \rightarrow \ |\ \boldsymbol{Z_{137MHz}}\ | \approx \mathbf{408\ \Omega}$$

With this shunt added to the circuit:

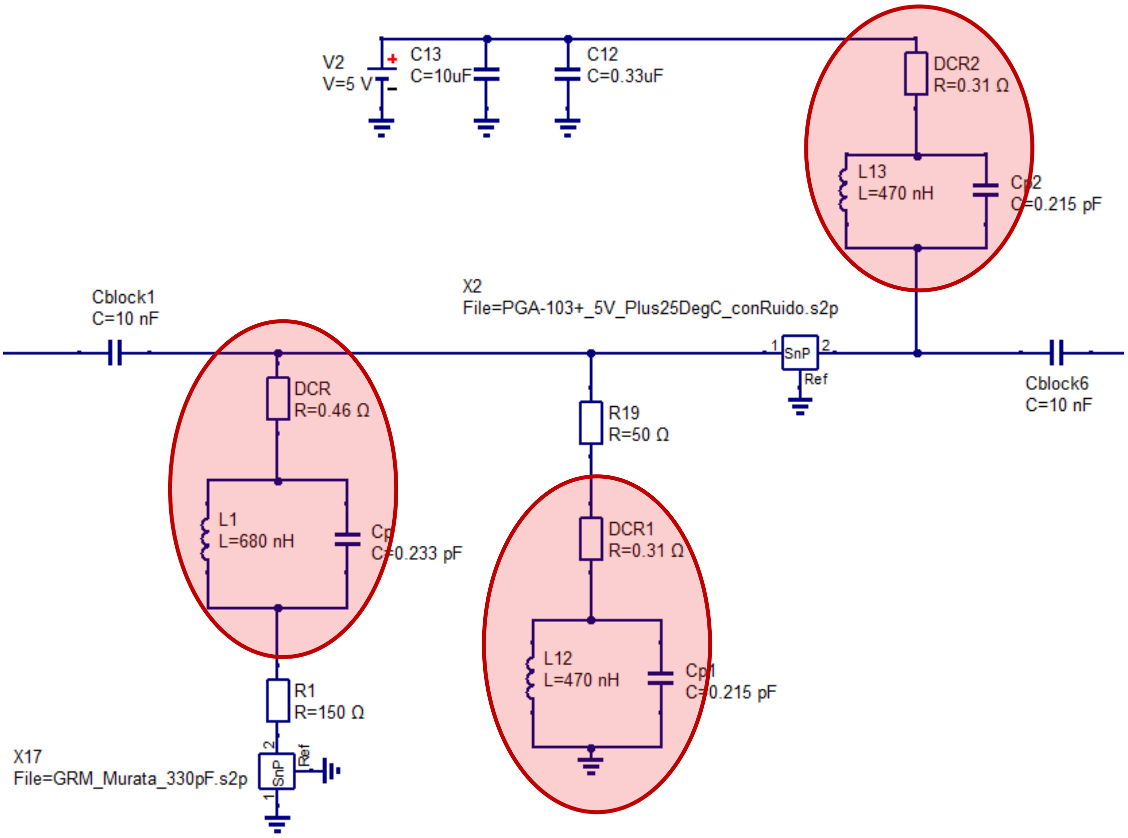


Circuit simulating to 240MHz adding the R + L shunt for stability.



The circuit now satisfies the stability criterion from 10MHz to 240MHz, while we already cover our operating frequency and low frequencies, we still must check stability across all the frequencies the LNA can function (10MHz to 4GHz).

I will be using the parasitic model for the 470nH and 680nH to simulate up to 1.7GHz:



$D = \det S(S) = 0.272 \,/\, -178°$

$\min(u) = 1.03$

$\min(K) = 1.07$

We can see the circuit is still stable up to 1.7GHz using the parasitic model.

If we simulate up to 4GHz we get the same results, these results up to 4GHz are **not** the real behavior of the circuit it's just an indicative.

I tried moving both my own stabilization branches, the PGA-103's own datasheet network and adding a resonant notch branch, nothing changes this specific dip of the µ-factor .

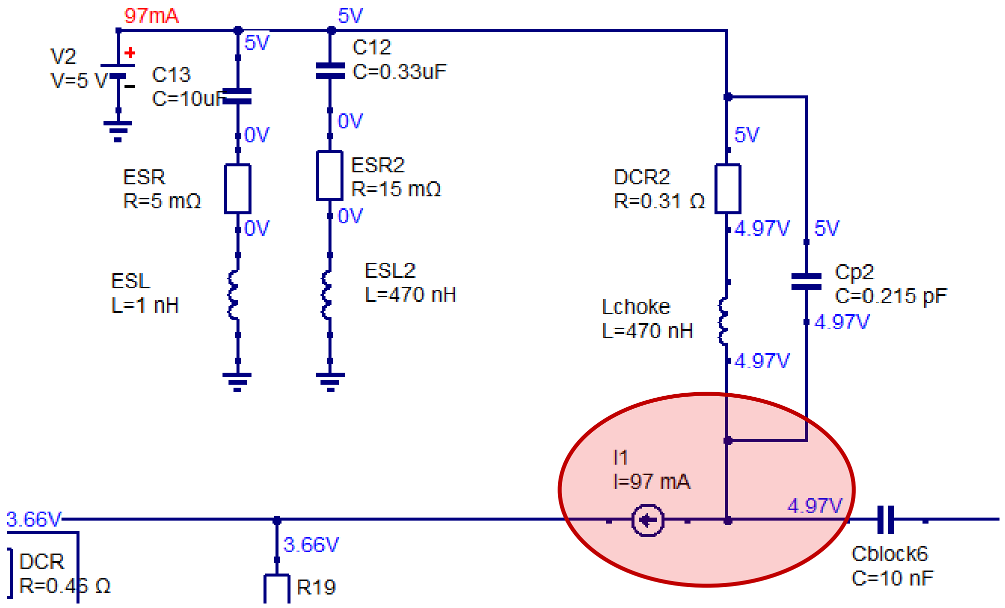
Looking at the bare device S-parameters alone (no external network at all), µ already sits at 0.81 at 50MHz and climbs gradually to 1.05 by 170MHz, so this is not caused by my design, it's an intrinsic property of the transistor itself in this frequency range.

This means my margin here is limited by the device physics, not by my component tolerances, so I've decided to accept this value instead of chasing a target that isn't achievable with this specific LNA.

I'm aware that a value of 1.07 for the K factor is dangerously close to the edge, but I want to add the 3rd stage to see if it will go up or drop below 1 before adding any additional shunt or network.

We still have to check that the LNA receives enough voltage and current to behave correctly in its linear zone, as I said before I already used components with low DCR.

I modeled the PGA using a DC current source with a value of 97mA:



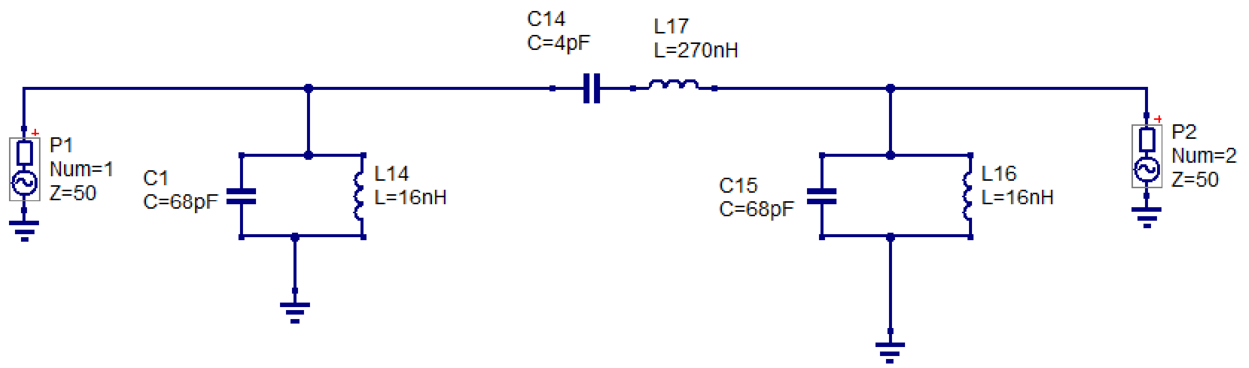
Bias tee modeling the LNA with a current source rather than the s2p file or an equivalent resistor (the impedance of the LNA varies with the frequency so using a fixed impedance would have been inaccurate)

Simulation shows 4.97 V at the PGA-103+ Vdd pin, which falls within the recommended 4.8 V–5.2 V window. The drop is consistent with $V_{drop} = I_{DD} \cdot DCR = 97 \text{ mA} \times 0.31\ \Omega = 30 \text{ mV}$, the DC biasing is correctly dimensioned.

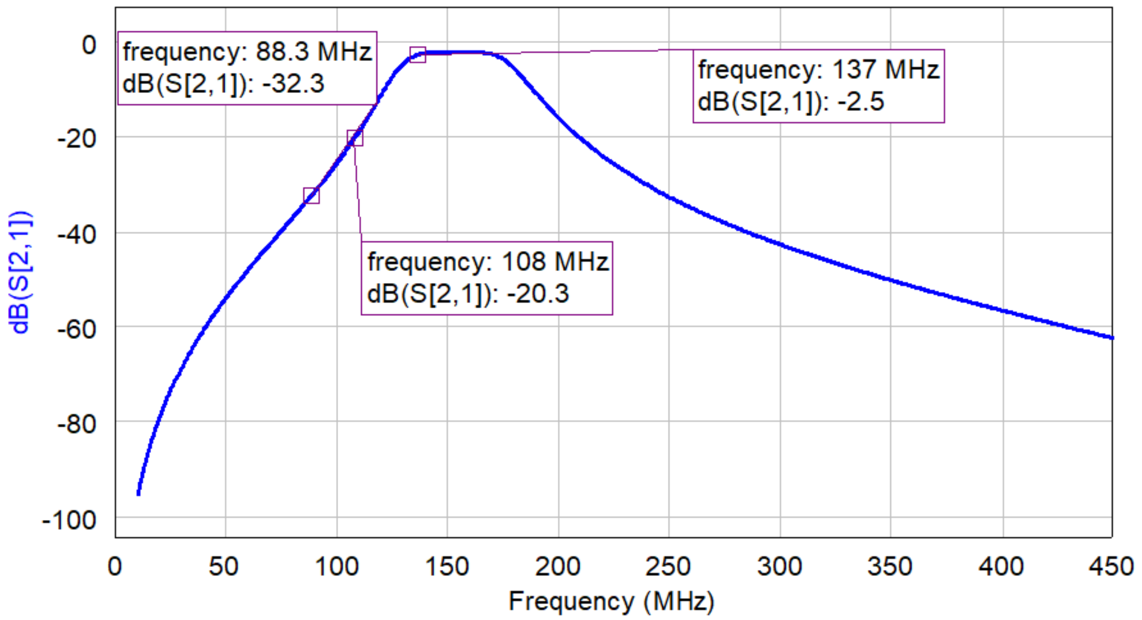
4. Final Stage: Passband 3rd order Butterworth.

With the LNA core and its stabilization networks finalized, I turned my attention to the output stage and the SDR itself. Feeding the amplified signal directly into a wideband receiver without adequate filtering can lead to significant performance degradation. SDRs are inherently susceptible to aliasing; any strong signal above half the sampling frequency (Fs/2) can fold back and fall directly within the 137 MHz target band. Furthermore, the LNA's gain, approximately 21 dB, amplifies not only the desired downlink, but also residual FM harmonics and out-of-band noise extending well into the GHz range, which may saturate the SDR's ADC input.

To address this, I decided to design a final passive Butterworth filter. I opted for this topology over Chebyshev or Cauer because its maximally flat passband response ensures that no amplitude ripple is introduced at 137 MHz, preserving signal integrity without phase or amplitude distortion. Its smooth, monotonic roll-off, starting around 200 MHz, provides effective attenuation of aliasing-prone frequencies and high-frequency noise. Placing this stage after the LNA was a deliberate choice: the insertion loss it contributes, approximately 1-2 dB, does not meaningfully degrade the overall system noise figure, as the preceding PGA-103+ gain comfortably masks this loss. This makes the Butterworth filter a suitable final step before delivering the signal to the SDR input.



Circuit design of the filter using the tool: "filter synthesis" from QUCS-S, the components have been changed for the .s2p files when simulating.



Looking at the simulated response, the Butterworth filter behaves exactly as expected. The FM band (88–108 MHz) is heavily attenuated—the signal barely gets through—so any residual FM that survived the Cauer notch is now effectively wiped out. The filter starts to roll off gently from around 185 MHz, reaching significant attenuation beyond 200 MHz, which is perfect for keeping harmonics and out-of-band noise away from the SDR. At 137 MHz, the insertion loss is moderate but because it is placed ***after*** the LNA the contribution to the overall NF is negligible.

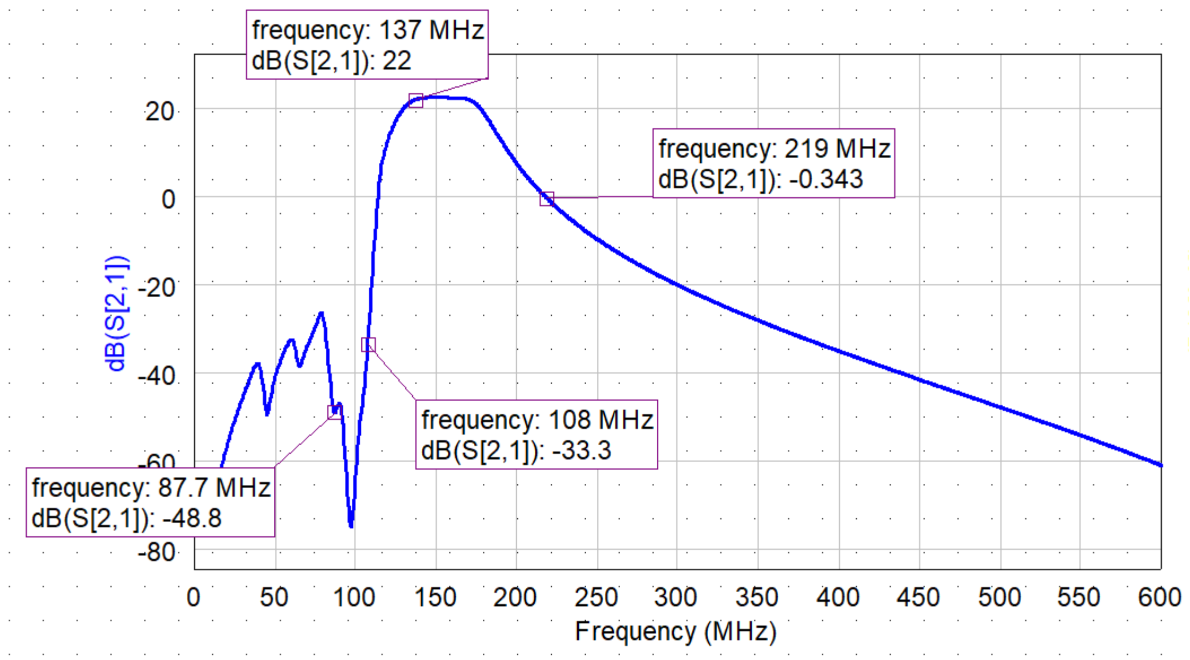
Quick note for the K factor, the value with this additional stage is 1.51. Meaning we are well above the edge, and we don't need any additional shunt nor network.

$$\min(u) = 1.04$$

$$\min(K) = 1.51$$

$$D = \det S(S) = 0.16 \,/\, 135^\circ$$

With all three stages cascaded, Cauer notch filter, LNA, and Butterworth output filter, I simulated the complete front-end response.



At 137 MHz, the cascade provides a clean **22dB** of gain, more than enough to drive the SDR while keeping a comfortable margin before compression. The out-of-band rejection is where this design truly shines. At 88.1 MHz the signal is attenuated to **-48.8 dB**, and at 108 MHz it sits at **-33.3 dB**. Even after the LNA amplifies everything, the FM residual remains deeply buried, ensuring the SDR won't suffer from intermodulation or saturation.

On the high-frequency side, the roll-off starts around 185 MHz and reaches unity gain (0 dB) at approximately 219 MHz. Beyond that, the Butterworth takes over and attenuation increases rapidly, eliminating any risk of aliasing.

In short, the cascade does exactly what I set out to achieve: amplify the desired signal with minimal noise, crush the FM broadcasters, and keep high-frequency garbage away from the ADC. The numbers confirm that the design meets its objectives with room to spare.

5. Overall Cascade Noise Analysis

With all three stages defined, the input Cauer notch filter, the PGA-103+ LNA, and the output Butterworth filter, I wanted to quantify the overall noise figure of the front-end. This is not just an academic exercise; it tells me whether my receiver will be sensitive enough to pick up the weak satellite signals or if it will be dominated by its own noise.

Using Friis' formula for a cascaded system:

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 \cdot G_2} + \cdots$$

where F and G are in linear units (not dB), I computed the contribution of each stage.

Stage	Component	Gain (dB)	NF (dB)
1	Cauer notch filter	**−1.09**	**1.09**
2	PGA-103+ LNA (with all biasing and stabilization networks)	**+22.6**	**0.732**
3	Butterworth pass-band filter	**−2.5**	**2.5**

Converting to linear factors:

- Stage 1: $G_1 = 10^{-1.09/10} = 0.778$, $F_1 = 10^{1.09/10} = 1.285$
- Stage 2: $G_2 = 10^{22.6/10} = 182.0$, $F_2 = 10^{0.732/10} = 1.184$
- Stage 3: $G_3 = 10^{-2.5/10} = 0.562$, $F_3 = 10^{2.5/10} = 1.778$

Plugging into Friis:

$$F_{total} = 1.285 + \frac{1.184 - 1}{0.778} + \frac{1.778 - 1}{0.778 \cdot 182.0}$$

$$F_{total} = 1.285 + 0.2365 + 0.00549 = 1.527 \text{ (linear)}$$

Finally, converting back to dB:

$$NF_{total} = 10 \cdot \log_{10}(1.527) \approx 1.84 \text{ dB}$$

The overall noise figure of the complete front-end is around **1.84 dB**.

With a system NF of 1.84 dB, I'm well below the typical galactic noise floor at VHF (~4 dB). This means my receiver will be limited by the sky noise, not by its own internal noise.

PCB and measures with VNA will follow soon.